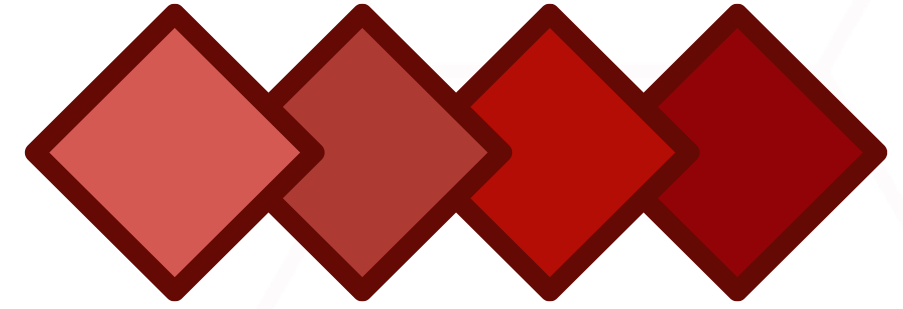
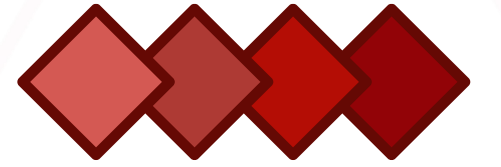
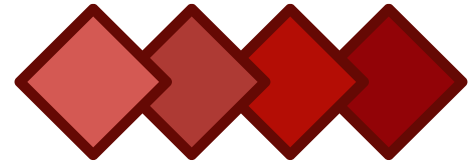




ORBITAL DECAY OF CLOSE-IN EXOPLANETS

Muhammad Areeb Qureshi

OVERVIEW



1. Intro and Motivation
2. Targets
3. Methodology
4. Main Results
5. Analysis
6. Conclusion and Further Work

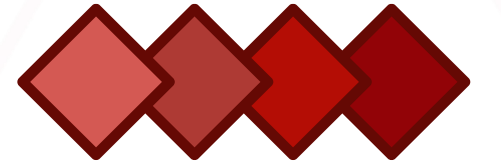
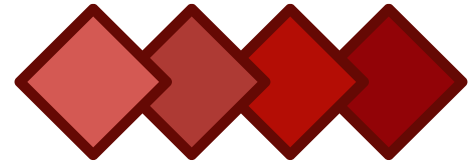
INTRO AND MOTIVATION

- Hot Jupiters are gas giant with masses and radii comparable to Jupiter, orbiting extremely close to their host stars.
- Tidal interactions in these systems cause the orbit to gradually shrink over time, leading to planetary engulfment.
- Measurements of orbital decay in hot Jupiter systems provide one of the only observational probes of tidal dissipation within stars, quantitatively characterised by the stellar tidal quality factor Q_* .

INTRO AND MOTIVATION

- Understanding the efficiency of tidal dissipation is essential for explaining the long-term evolution and fate of close-in planetary systems.
- However, measurement of Q'_* proves difficult as the orbital decay signal can be produced by alternative mechanisms such as apsidal precession or the Late Travel-Time Effect (LTTE).

KEY QUANTITY TO MEASURE



Tidal Quality Factor:

$$Q'_* = -\frac{27\pi}{2\dot{P}} \left(\frac{M_p}{M_*} \right) \left(\frac{R_*}{a} \right)^5 \left(\frac{\omega_*}{n} - 1 \right)$$

Where $\dot{P}$ is the orbital decay rate

M_* is the stellar mass

M_p is the planetary mass

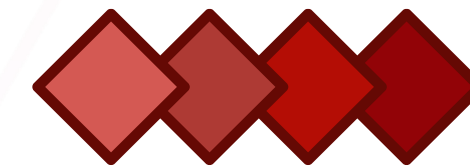
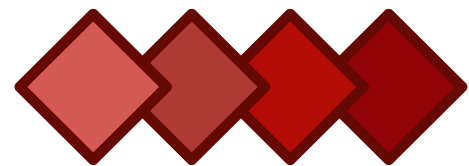
R_* is the stellar radius

a is the orbital separation

ω_*/n is the stellar rotational and orbital angular frequency

Ref. P. Goldreich and S. Soter, Icarus, 5, 375 (1966) and I. Wong et al., The Astronomical Journal 163, 175 (2022).

TARGETS



	Kepler-1658b	WASP-12b
P	3.85 d	1.09 d
M_p	$5.88 \pm 0.47 M_J$	$1.47 \pm 0.076 M_J$
R_p	$1.07 \pm 0.05 R_J$	$1.900 \pm 0.057 M_J$
Reason for selection	Controversial candidate, high tidal interactions expected	Benchmark model, only planet with confirmed orbital decay

Table 1. Comparison of the two targets

METHODOLOGY

Orbital decay can be measured using the transit method: detecting the periodic decrease in stellar brightness as a planet passes in front of its host star.

Key Parameter: mid-transit time t_0

The transit method:

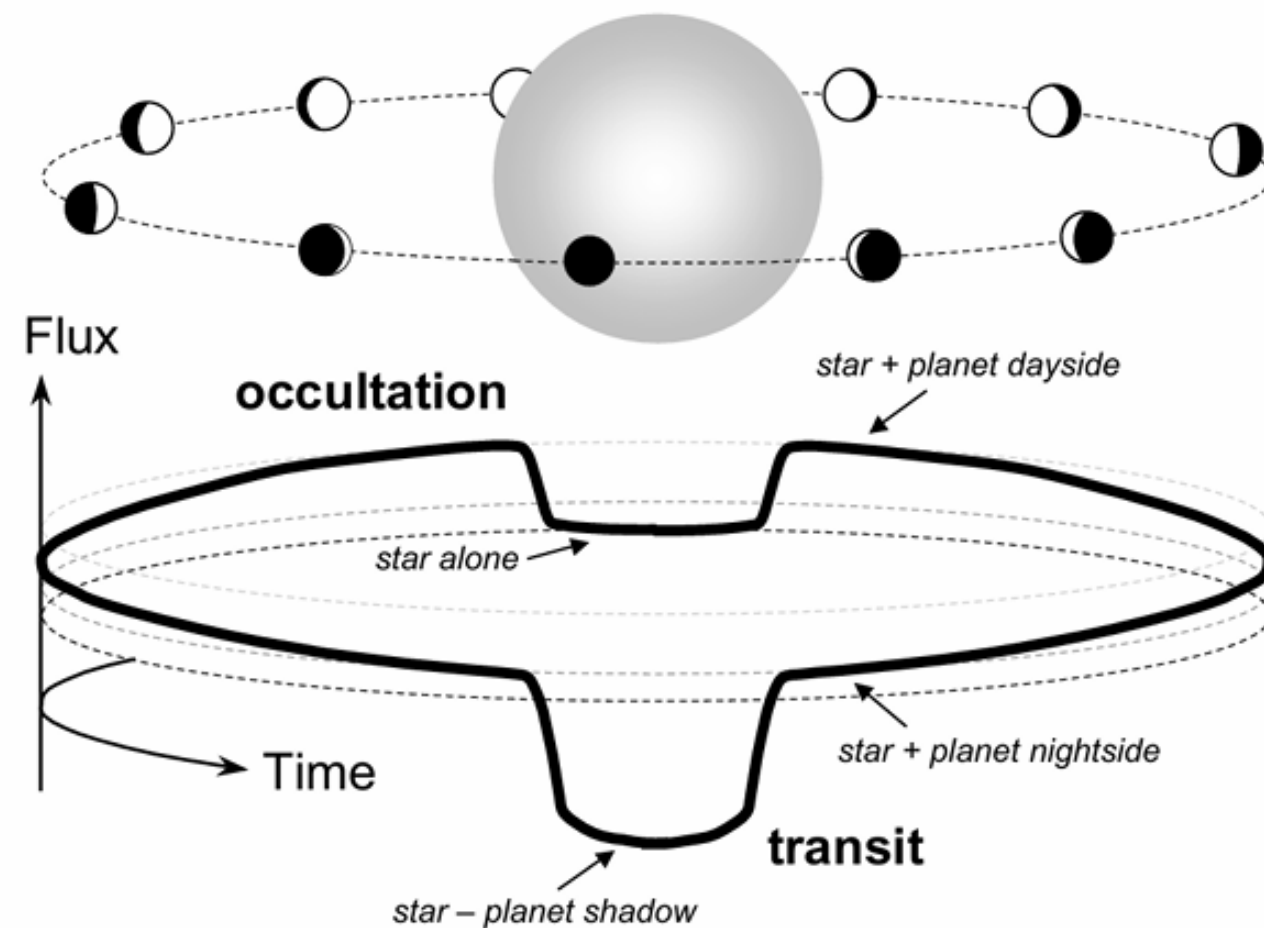
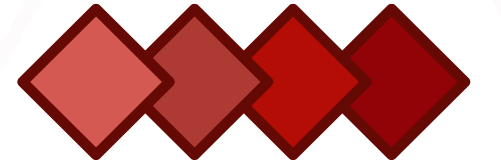
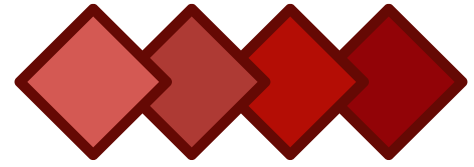


Fig. 1 The transit method

Some important parameters:

- 1) Impact parameter b (or inclination)
- 2) Stellar Limb-darkening
- 3) Ratio of planetary and stellar radii
- 4) Stellar density (or orbital separation)

METHODOLOGY



- Step 0:** Get the raw data
- Step 1:** Flatten and Bin the Data
- Step 2:** Fit a model based on priors
- Step 3:** Extract the posteriors
- Step 4:** Plot the O-C diagram and fit the models

Packages used:

- Juliet
- Lightkurve
- Corner.py
- Matplotlib
- Numpy
- Astropy (limited)

METHODOLOGY

Step 0: Get the raw data

- Download light curves from an archive (such as the MAST archive)
- Mask the primary transit points (using the period and duration of transits)

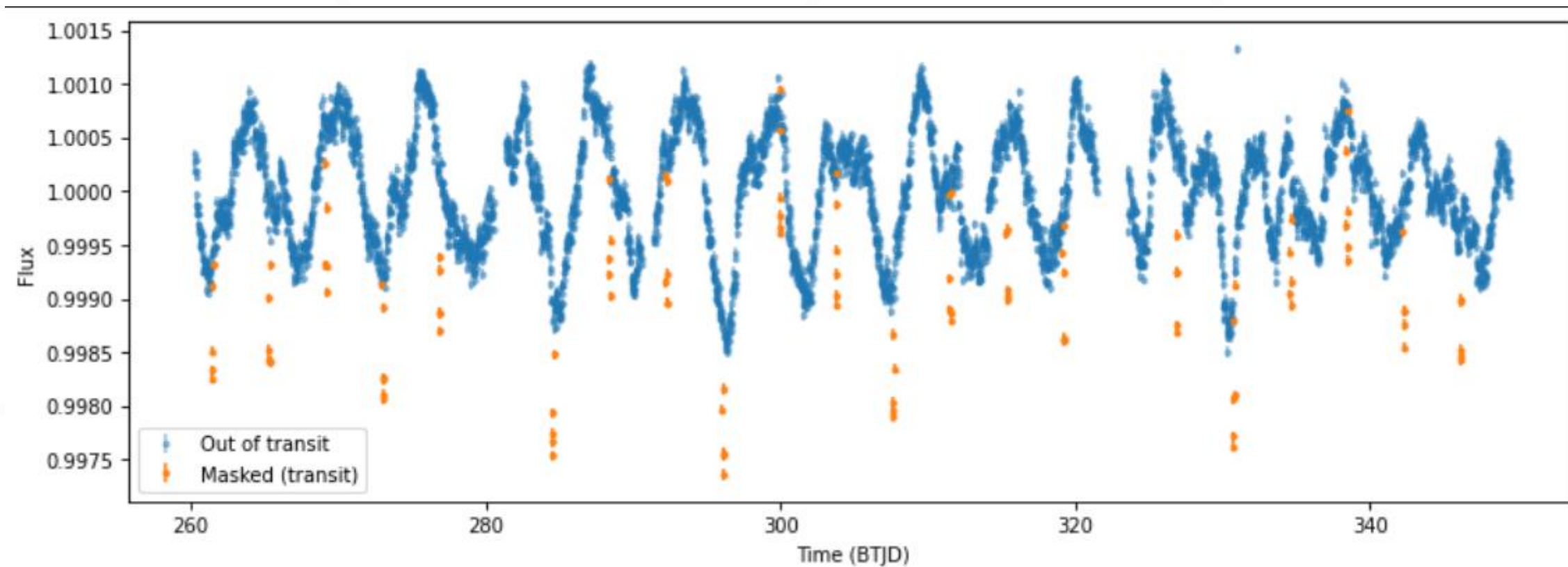


Fig. 2 Raw transit light curve data

METHODOLOGY

Step 1: Flatten and Bin the data

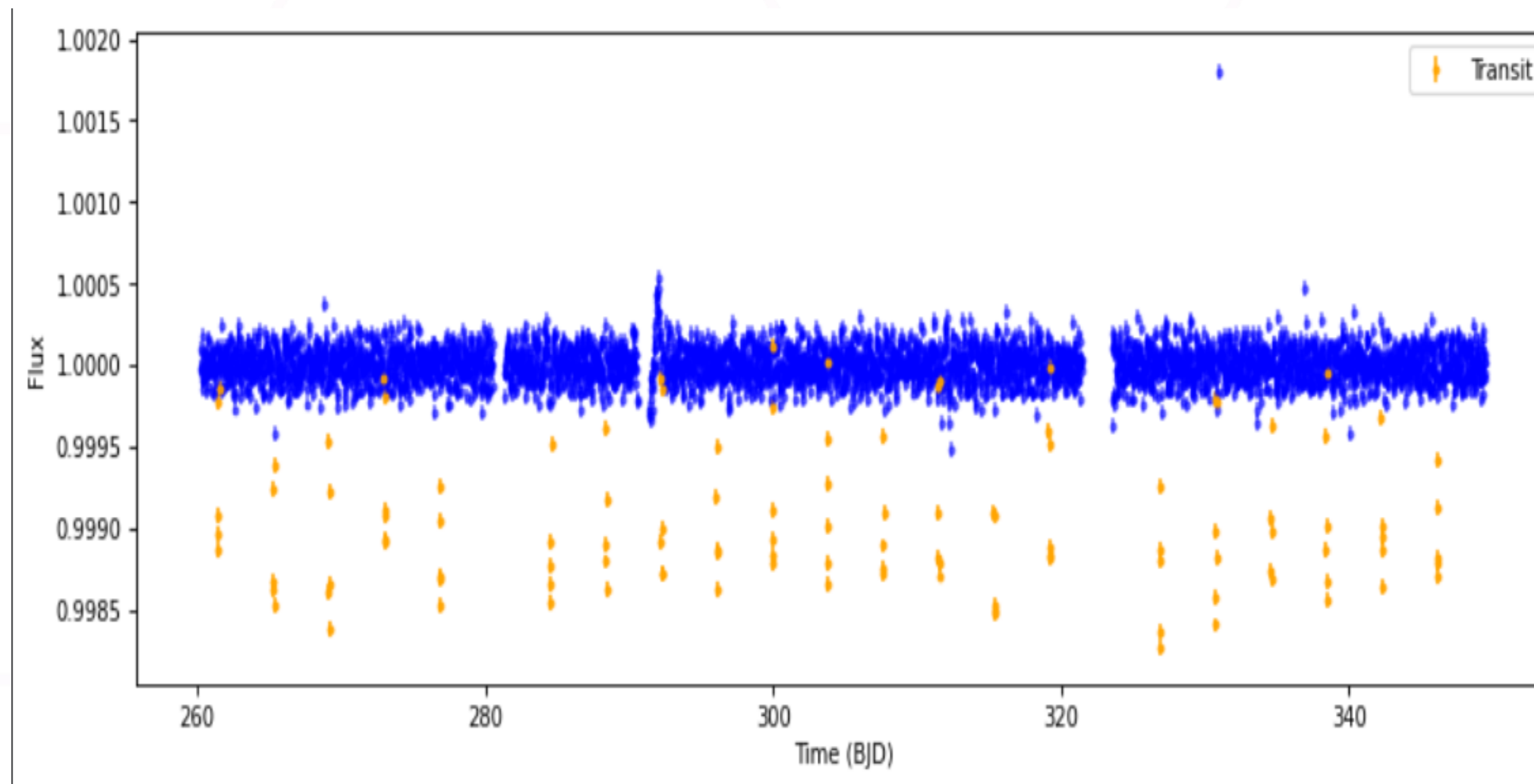


Fig. 3 Flattened and binned data

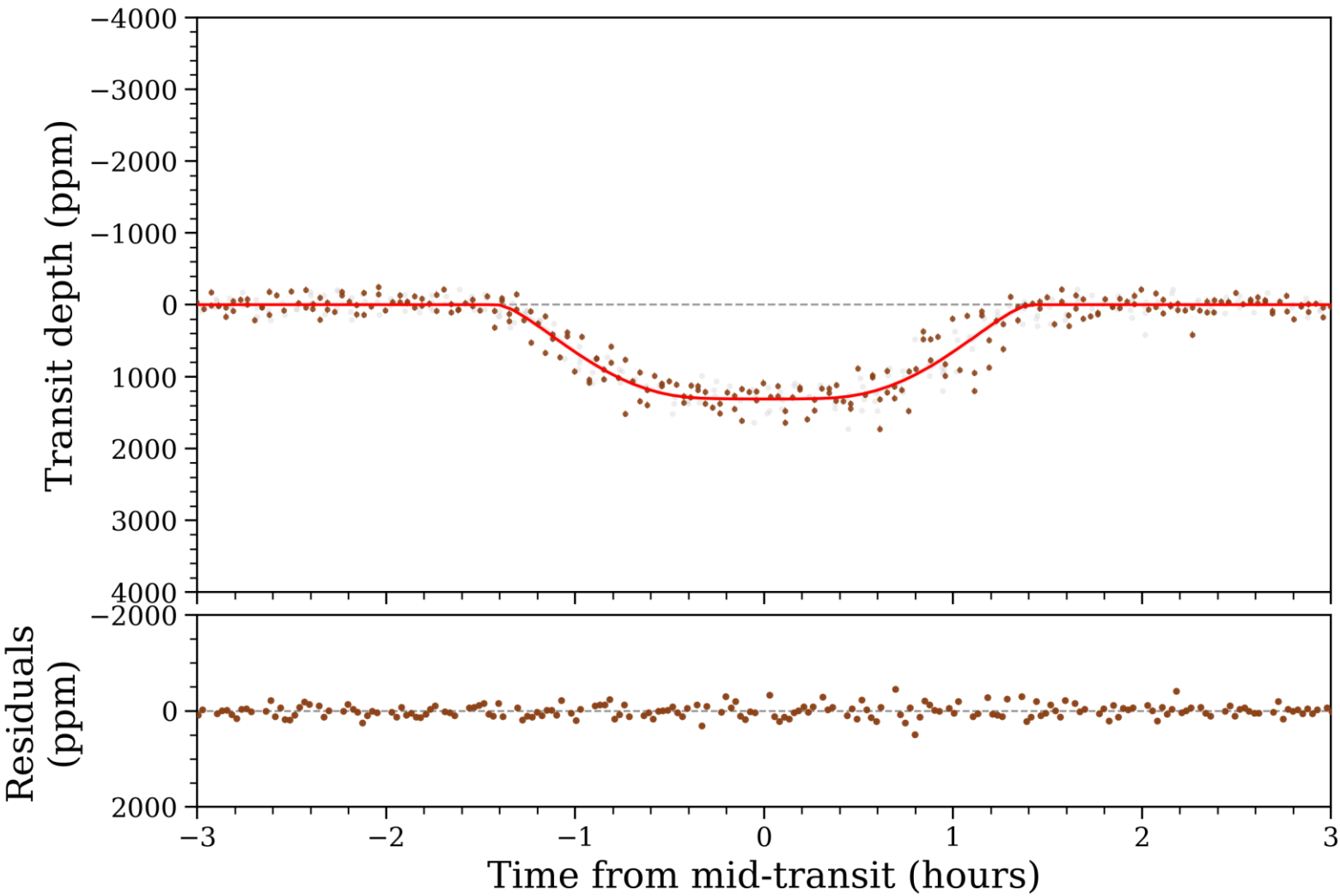
METHODOLOGY

Step 2: Fit a model based on priors

Parameter	Priors	Initial Point
P (days)	$U[3.75; 3.95]$	3.849
b	$U[0; 1]$	0.9502
$R_p/R_\star$	$U[0.02; 0.06]$	0.0359
$\rho_\star$ (kg m ⁻³)	$N(83.4; 7.9)$	106.8
q_1	$U[0; 1]$	0.3
q_2	$U[0; 1]$	0.3
e	0.0	-
ω	90.0	-
mdilution	1.0	-
mflux	$N(0; 0.1)$	0.0
σ_ω	$\log U[1; 1000]$	10

Table 2. Priors for Kepler-1658b

Fig. 4 MCMC transit model fit for a sample Kepler-1658b light curve



METHODOLOGY

Step 3: Extract the posteriors

Parameter	Kepler	TESS
P (days)	$3.849^{+1.4 \times 10^{-6}}_{-1.5 \times 10^{-6}}$	$3.849^{+9.4 \times 10^{-6}}_{-9.9 \times 10^{-6}}$
b	$0.946^{+0.004}_{-0.004}$	$0.962^{+0.007}_{-0.006}$
t_0	$*650.246^{+0.0001}_{-0.0001}$	$*3325.74^{+0.0011}_{-0.0010}$
$R_p/R_\star$	$0.0370^{+0.0007}_{-0.0008}$	$0.0440^{+0.005}_{-0.003}$
$\rho_\star$ (kg m $^{-3}$)	$136.8^{+8.214}_{-9.274}$	$88.89^{+7.50}_{-7.46}$
$a/R_\star$	$4.750^{+0.0932}_{-0.1099}$	$4.114^{+0.113}_{-0.119}$
q_1	$0.0872^{+0.0716}_{-0.0612}$	$0.114^{+0.203}_{-0.088}$
q_2	$0.101^{+0.201}_{-0.077}$	$0.469^{+0.362}_{-0.322}$

Table 3. Posteriors for Kepler-1658b

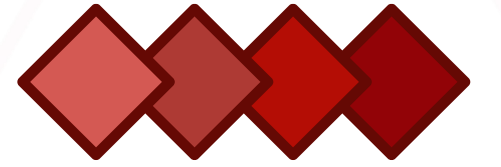
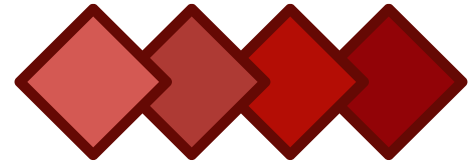
METHODOLOGY

Ref. S. Vissapragada et al., The Astrophysical Journal Letters 941, L31 (2022).

Data Set	Transit Time (BJD _{TDB})	This work
Kepler LC Quarter 0	2454959.7314 ^{+0.0014} _{-0.0015}	2454959.74007 ^{+0.00264} _{-0.00207}
Kepler LC Quarter 1	2454982.82835 ^{+0.00061} _{-0.00061}	2454982.82992 ^{+0.00138} _{-0.00094}
Kepler SC Quarter 2	2455048.26751 ^{+0.00021} _{-0.00022}	2455048.26818 ^{+0.00015} _{-0.00015}
Kepler LC Quarter 3	2455140.65189 ^{+0.00040} _{-0.00042}	2455136.80312 ^{+0.00054} _{-0.00086}
Kepler LC Quarter 4	2455233.03736 ^{+0.00033} _{-0.00035}	2455229.19019 ^{+0.00042} _{-0.00042}
Kepler LC Quarter 5	2455325.42133 ^{+0.00035} _{-0.00036}	2455325.42081 ^{+0.00044} _{-0.00049}
Kepler SC Quarter 7	2455510.19192 ^{+0.00023} _{-0.00023}	2455506.34276 ^{+0.00016} _{-0.00015}
Kepler SC Quarter 8	2455602.57708 ^{+0.00027} _{-0.00027}	2455602.57651 ^{+0.00018} _{-0.00018}
Kepler LC Quarter 9	2455691.11211 ^{+0.00033} _{-0.00031}	2455691.11243 ^{+0.00064} _{-0.00059}
Kepler LC Quarter 11	2455883.58121 ^{+0.00032} _{-0.00033}	2455879.73287 ^{+0.00041} _{-0.00043}
Kepler LC Quarter 12	2455975.96583 ^{+0.00034} _{-0.00036}	2455972.11824 ^{+0.00034} _{-0.00072}
Kepler LC Quarter 13	2456064.50087 ^{+0.00036} _{-0.00036}	2456060.65134 ^{+0.00056} _{-0.00050}
Kepler LC Quarter 15	2456256.97026 ^{+0.00037} _{-0.00035}	2456256.97027 ^{+0.00041} _{-0.00045}
Kepler LC Quarter 16	2456349.35438 ^{+0.00038} _{-0.00039}	2456353.20533 ^{+0.01219} _{-0.00130}
Kepler LC Quarter 17	2456410.94385 ^{+0.00065} _{-0.00064}	2456407.09430 ^{+0.00082} _{-0.00096}
Palomar/WIRC Visit 1	2459097.8002 ^{+0.0015} _{-0.0015}	-
TESS Sector SC 41	2459436.5407 ^{+0.0023} _{-0.0023}	2459432.69245 ^{+0.00281} _{-0.00287}
Palomar/WIRC Visit 2	2459790.6819 ^{+0.0015} _{-0.0013}	-
TESS Sector SC 54	2459786.8359 ^{+0.0028} _{-0.0030}	2459782.98873 ^{+0.00362} _{-0.00352}
TESS Sector SC 55	2459813.7791 ^{+0.0029} _{-0.0027}	2459809.92771 ^{+0.00295} _{-0.00296}
TESS Sector FC 74	-	2460325.74092 ^{+0.00298} _{-0.00292}
TESS Sector FC 75	-	2460352.68851 ^{+0.00267} _{-0.00267}
TESS Sector FC 81	-	2460522.05581 ^{+0.00280} _{-0.00273}
TESS Sector FC 82	-	2460545.15622 ^{+0.00298} _{-0.00305}

Table 4. Comparison between mid-transit times from literature and this work

METHODOLOGY



Step 4: Plot the O-C diagram and fit the models

$$t(N) = t_0 + NP \text{ (constant period)}$$

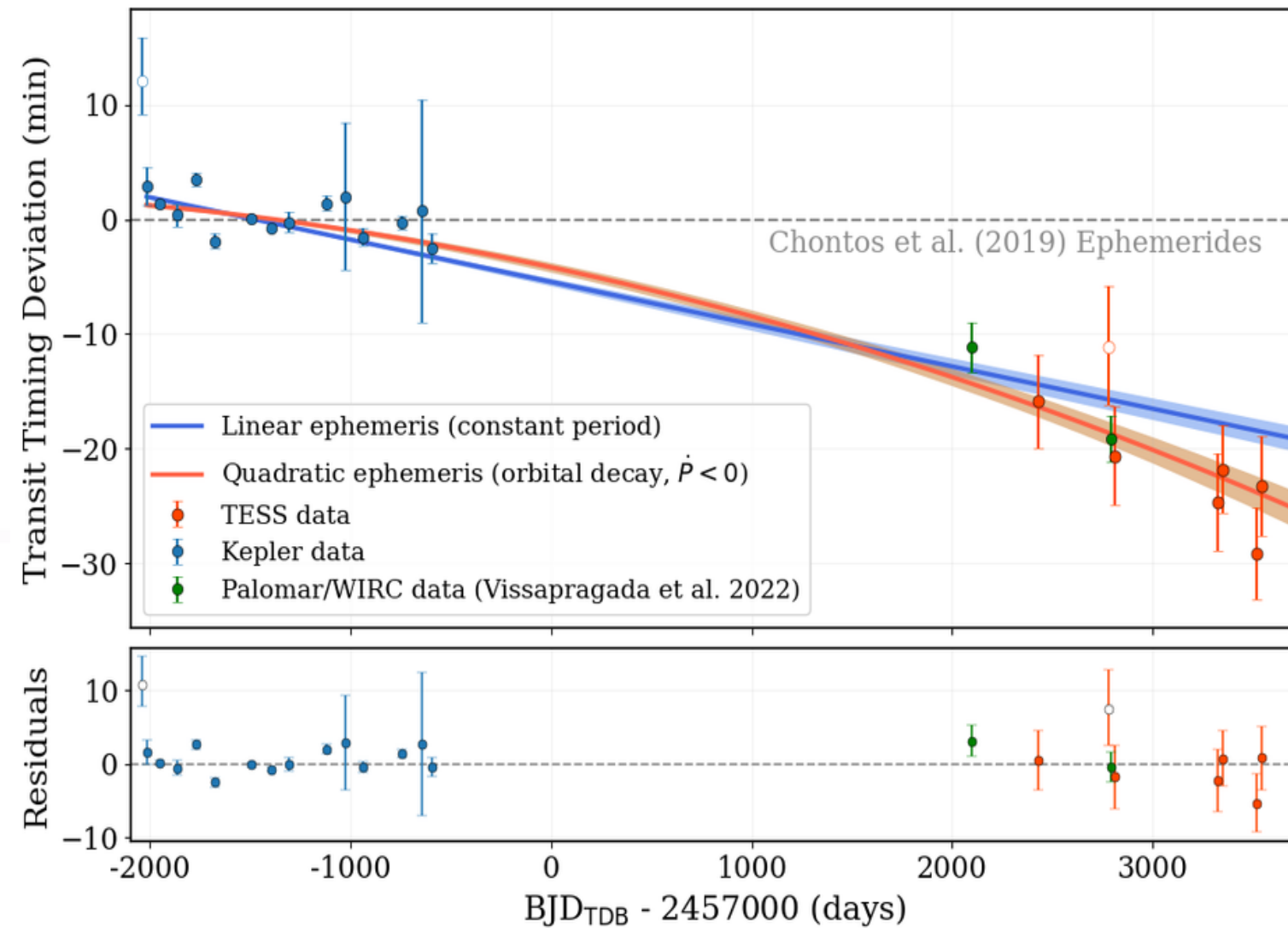
$$t(N) = t_0 + NP + \frac{1}{2} \frac{dP}{dN} N^2 \text{ (orbital decay)}$$

Where $\frac{dP}{dN} = \dot{P}$ and $\tau = \frac{P}{|\dot{P}|}$

Ref. S. Vissapragada et al., The Astrophysical Journal Letters 941, L31 (2022).

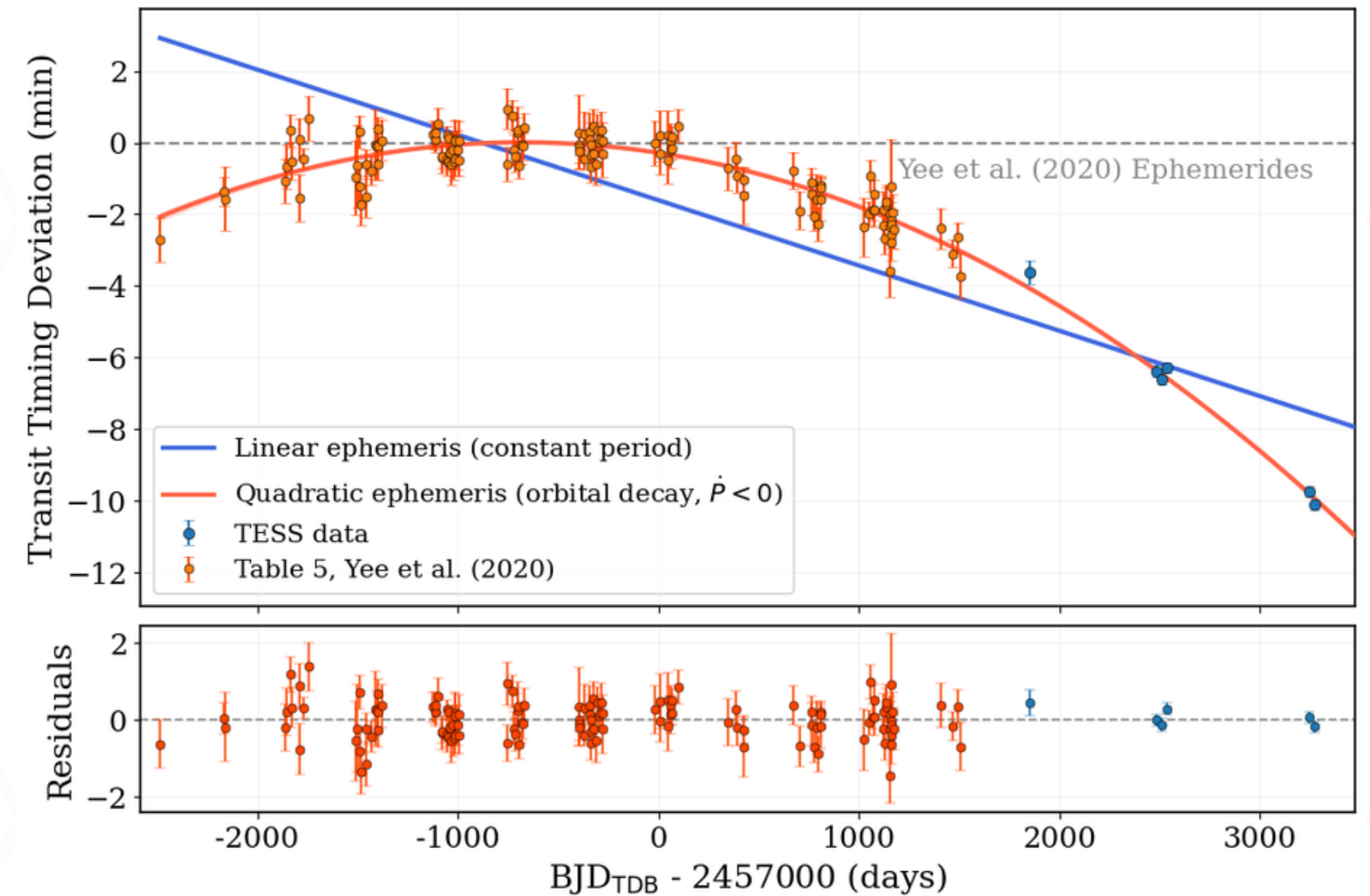
RESULTS

Fig. 5 Kepler-1658b O-C Plot



Statistically favoured orbital decay model

Fig. 6 WASP-12b O-C Plot



Unambiguous orbital decay model

after extending the baseline with previously unexplored datasets

RESULTS

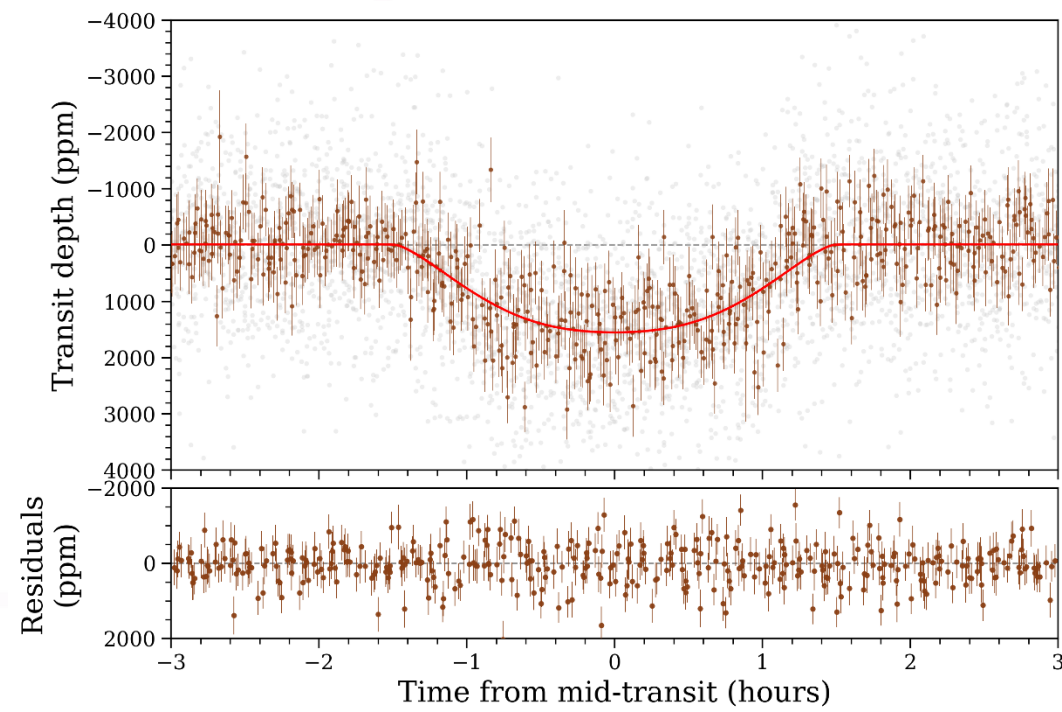
Parameter	WASP-12b		Kepler-1658b	
	Constant	Decay	Constant	Decay
P (days)	1.091418652(16)	1.091420116(45)	3.849363450(56)	3.849368844(119)
t_0	2456305.455425(27)	2456305.455811(29)	2455005.925460(107)	2455005.924991(140)
$\dot{P}$ (ms yr ⁻¹)	–	$-30.298^{+0.857}_{-0.859}$	–	$-87.853^{+16.846}_{-16.816}$
τ (Myr)	–	$3.112^{+0.091}_{-0.086}$	–	$3.786^{+0.898}_{-0.608}$
Q'_*	–	$(1.820^{+0.195}_{-0.173}) \times 10^5$	–	$(1.842^{+0.484}_{-0.681}) \times 10^4$
<i>Fit Statistics</i>				
N_{dof}	105	104	20	19
χ^2_{min}	1356.619	116.445	92.865	66.426
AIC	1360.619	122.445	96.865	72.426
BIC	1365.965	130.464	99.047	75.699

Derived Parameters	WASP-12b	Wong et al. 2022 [4]	Kepler-1658b	Vissapragada et al. 2022) [5]
$\dot{P}$ (ms/yr)	-30.3 ± 0.9	-29.81 ± 0.94	-87.9 ± 16.8	-131 ± 20
τ (Myr)	3.11 ± 0.09	3.16 ± 0.10	3.79 ± 0.7	≈ 2.5
Q'_*	$(1.82 \pm 0.18) \times 10^5$	$(1.50 \pm 0.11) \times 10^5$	$(1.84 \pm 0.5) \times 10^4$	$(2.50 \pm 0.7) \times 10^4$
ΔBIC	1235.5	–	23.4	–
χ^2	1.12	–	3.50	–

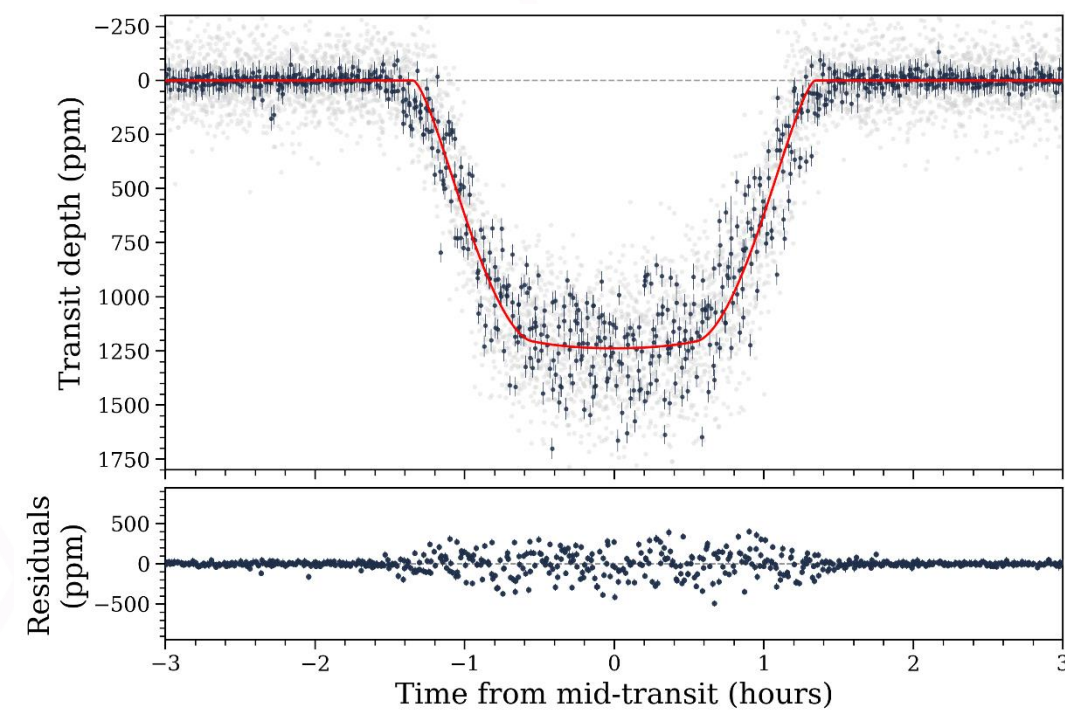
Table 4 (top). Derived parameters for the two planets with statistical analysis

Table 5 (bottom) Comparison with literature values

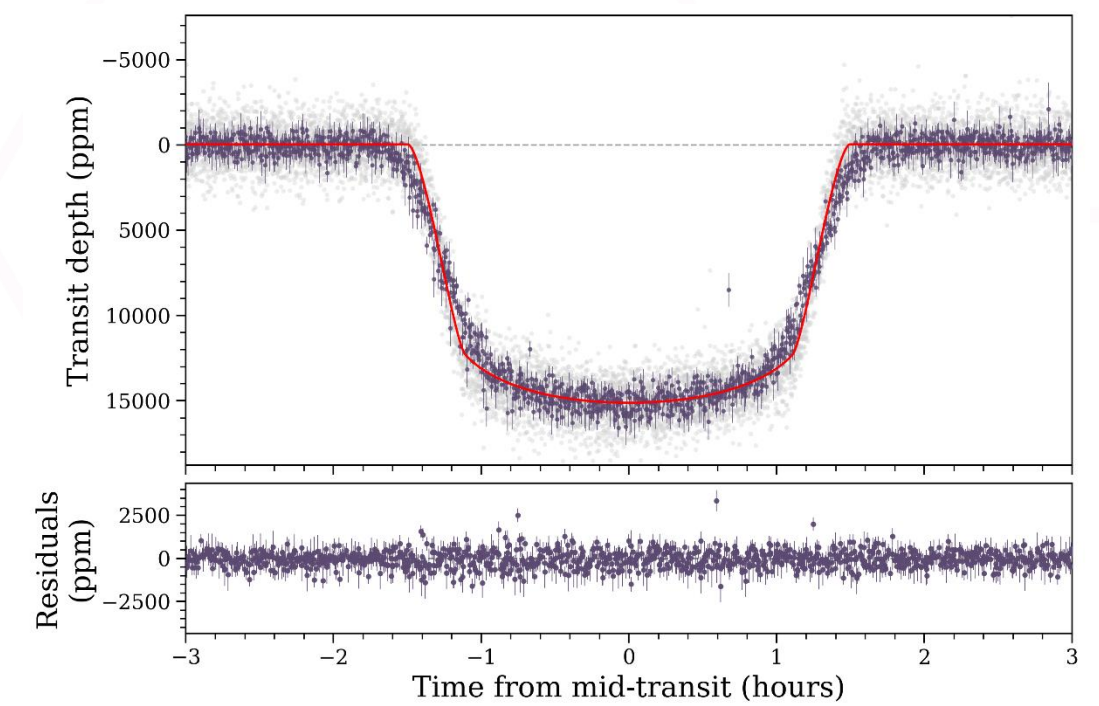
RESULTS



**Fig. 7 Kepler-1658b
(TESS) global plot**



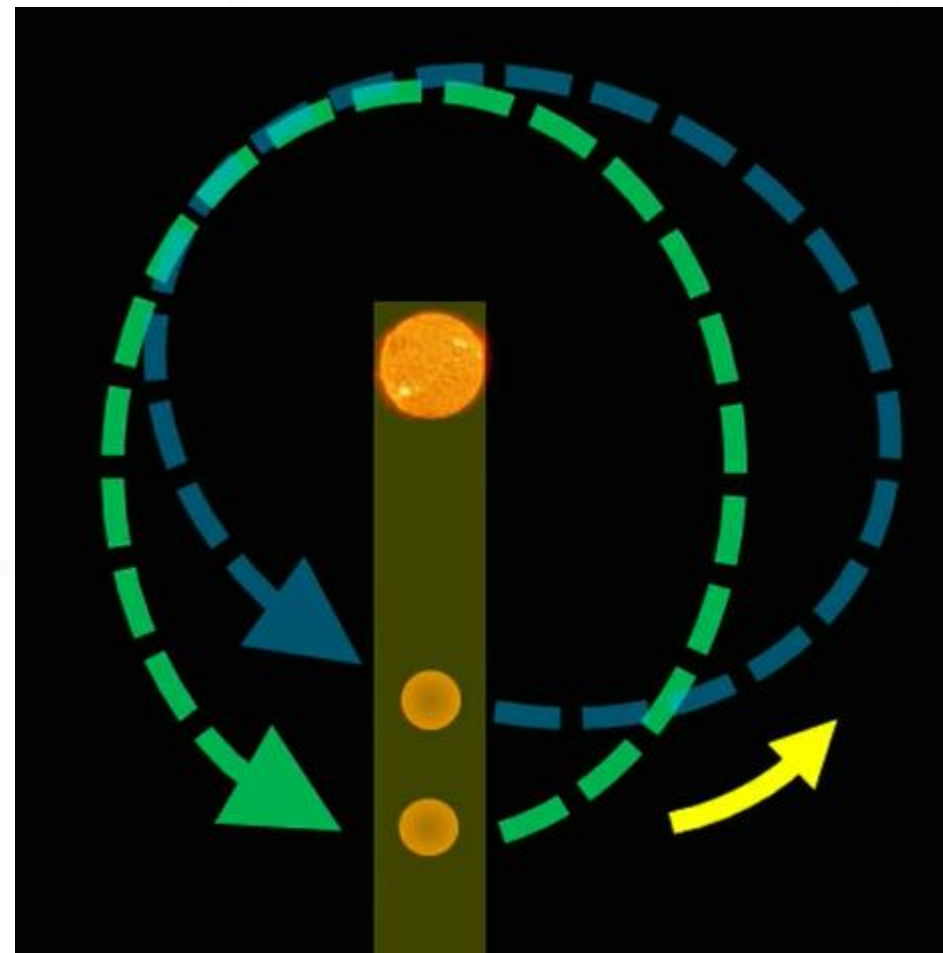
**Fig 8. Kepler-1658b
(Kepler) global plot**



**Fig. 9 WASP-12b global
plot**

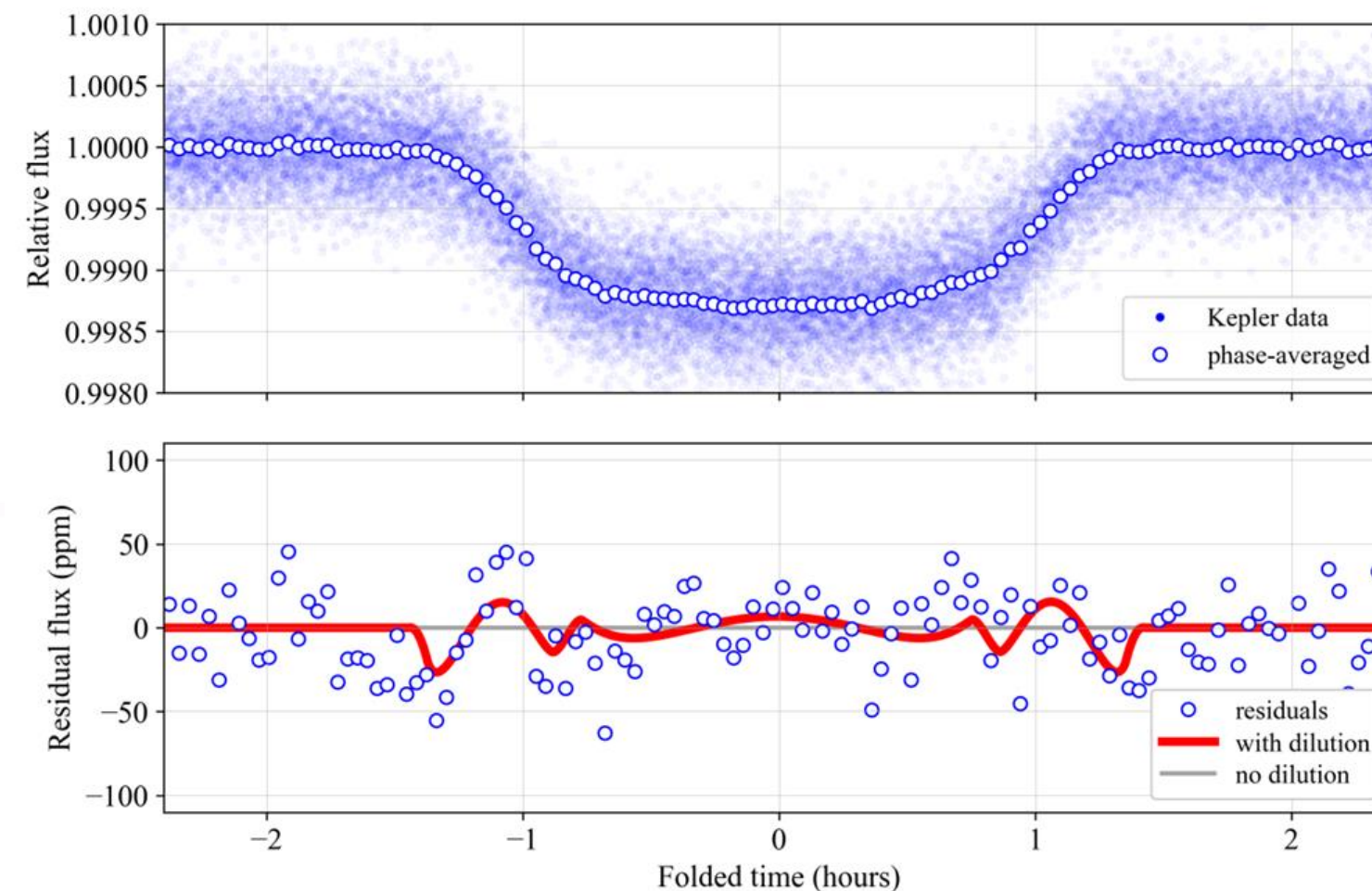
ALTERNATIVE MECHANISMS

Apsidal Precession



Resolved by checking
secondary eclipses

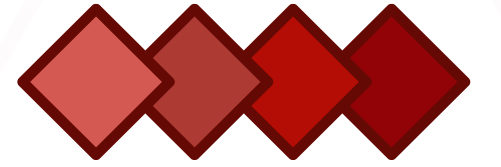
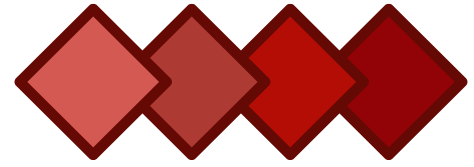
Light travel-time effect



Resolved by doing detailed
RV analyses

Ref. J. N. Winn and G. Stef'ansson, (2025),
arXiv:2510.05229

DISCUSSION



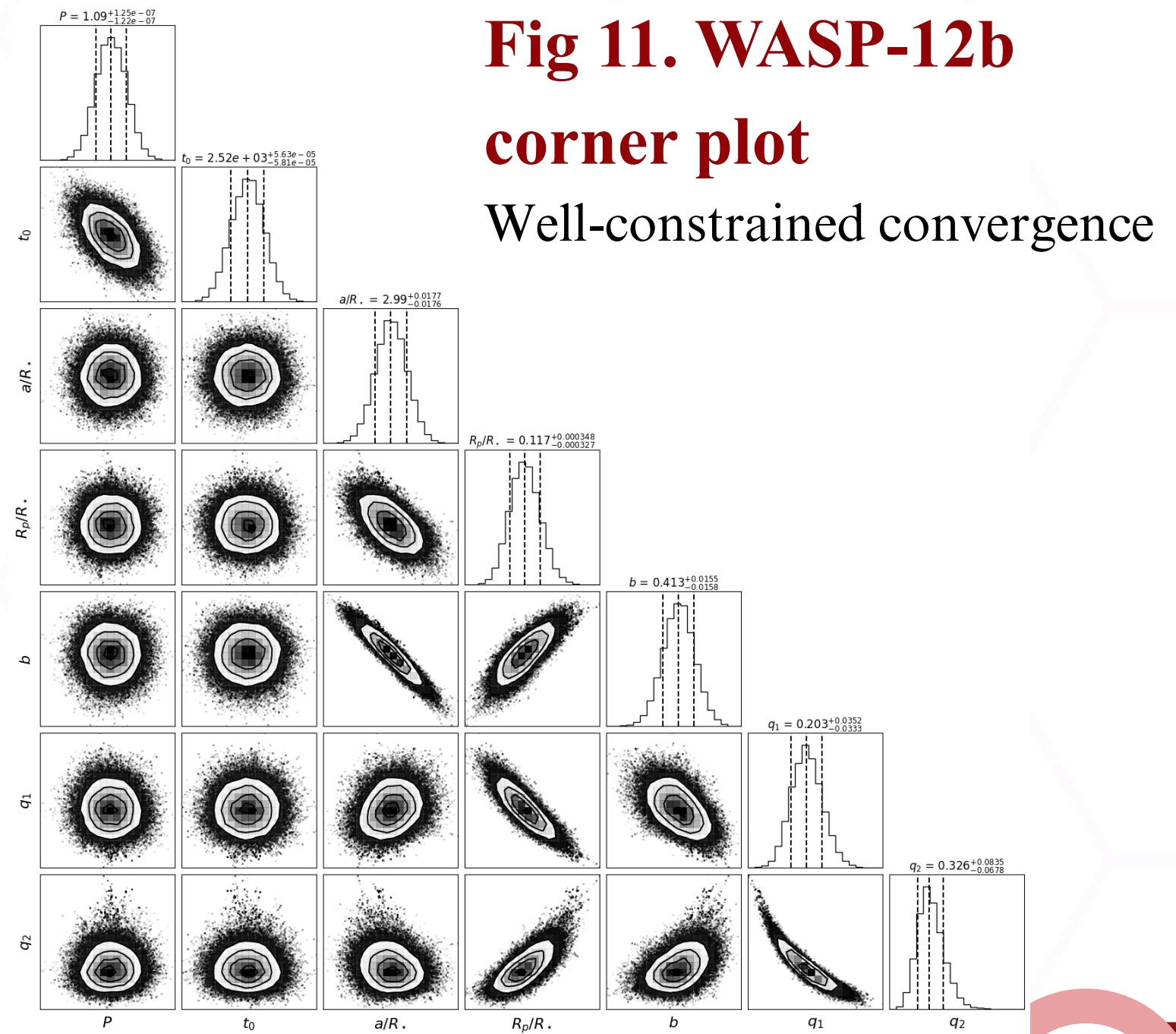
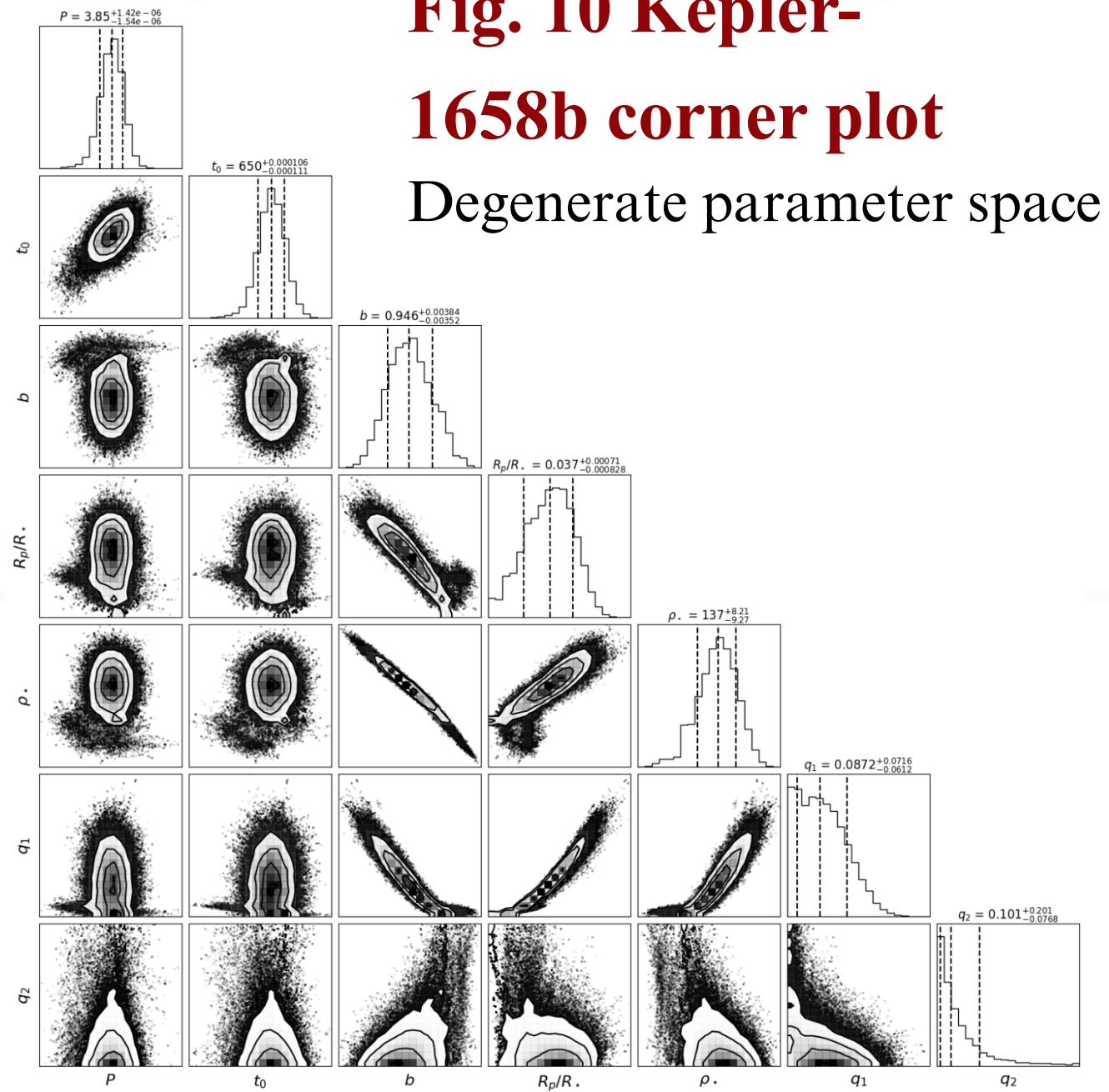
1. WASP-12b

Unambiguous results consistent with literature values, no sign of apsidal precession or LTTE. Only confirmed example of orbital decay. $\dot{P}$, τ and Q_* in excellent agreement with previous studies.

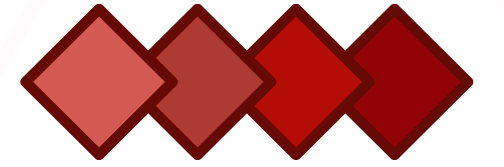
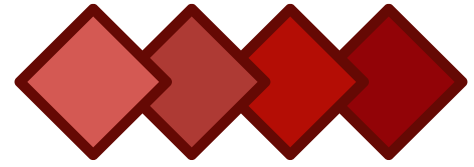
2. Kepler-1658b

Statistically favoured orbital decay model with no signs of apsidal precession; however, strong signals of LTTE from a nearby companion at 50AU. RV analysis and density estimates suggest a triple-star model.

SOME OTHER ANALYSIS



CONCLUSION AND FURTHER WORK



- WASP-12b remains the only known and confirmed Hot Jupiter exhibiting tidal orbital decay.
- Kepler-1658b may not even be a planet.
- Tidal theory remains poorly constrained.
- Primary transit analysis by itself is insufficient to confirm orbital decay.
- Work very well received by department and encouraged to publish the report. Scored one of the highest marks in cohort (78%).
- Studies can be extended by joint radial velocity (RV) and secondary eclipse analyses.
- Future projects such as PLATO, NEID and data from JWST may help with that.

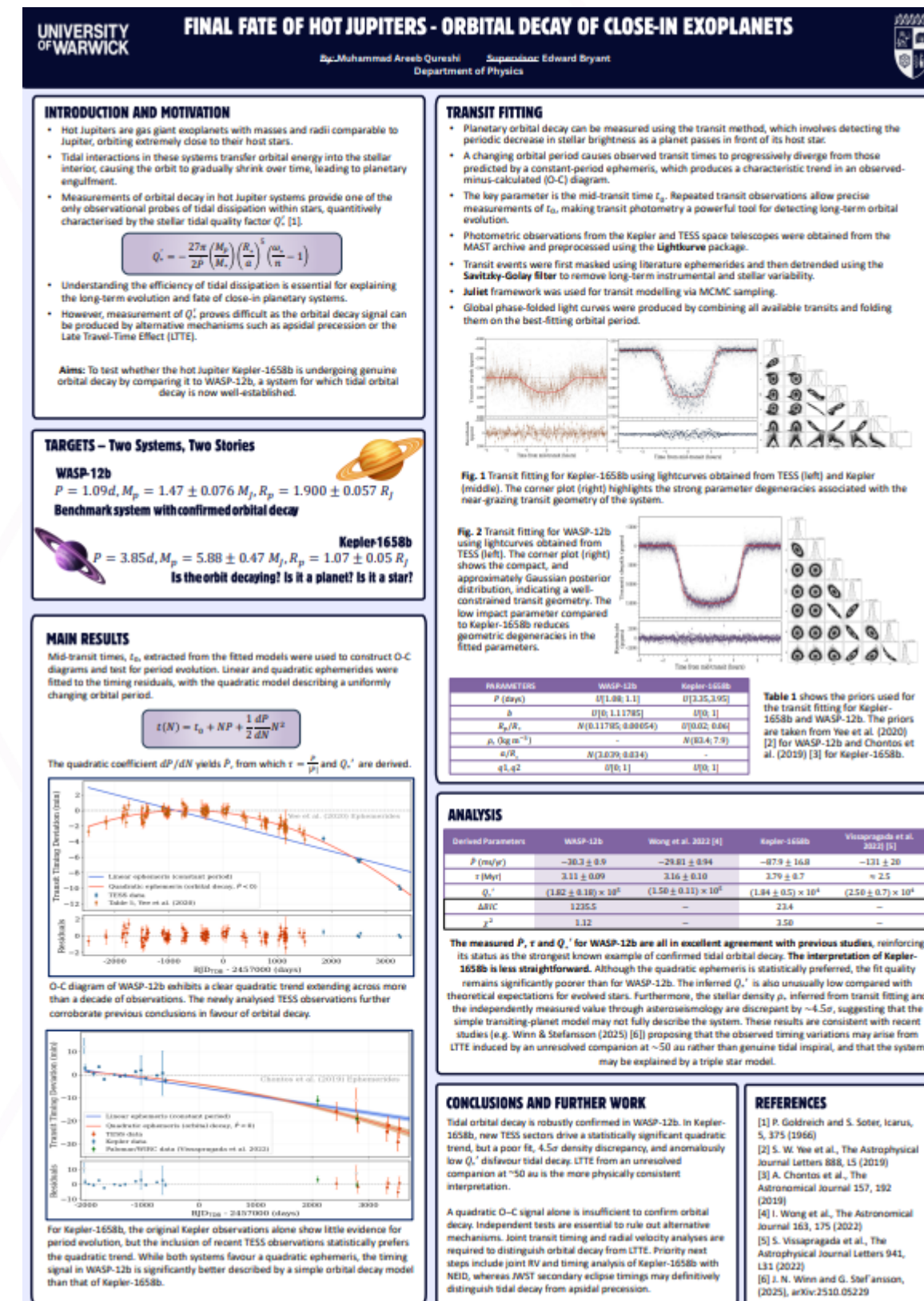
FINAL COURSEWORK

Orbital Decay of Close-in Exoplanets - Final Report

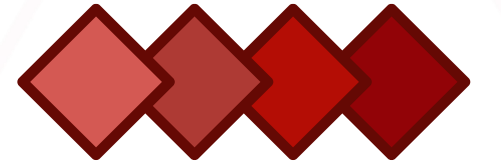
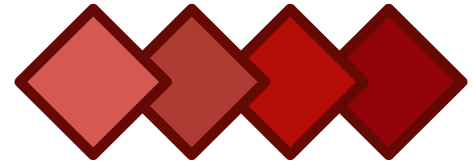
2210176

Department of Physics, University of Warwick, Coventry CV4 7AL, United Kingdom

Tidal orbital decay of hot Jupiters offers a rare probe of tidal dissipation in stellar interiors. We analyse transit timing variations of two hot Jupiter systems, WASP-12b and Kepler-1658b, using Kepler and TESS photometry, modelling light curves with an MCMC framework, and constructing O-C diagrams to test for period evolution. For WASP-12b, we confirm an unambiguous quadratic trend with tidal decay rate $\dot{P} = -30.298^{+0.857}_{-0.859}$ ms yr⁻¹ and a stellar tidal quality factor $Q'_\star = (1.820^{+0.195}_{-0.173}) \times 10^5$ using new TESS observations, reinforcing its status as the best-established case of tidal inspiral. For Kepler-1658b, we detect a significant quadratic trend with $\dot{P} = -87.853^{+16.846}_{-16.816}$ ms yr⁻¹ and $Q'_\star = (1.842^{+0.484}_{-0.681}) \times 10^4$, driven primarily by new TESS sectors 74, 75, 81, and 82 not previously analysed. However, inconsistencies including poor model fits, discrepancies in stellar densities obtained from photometric and asteroseismic models and anomalously low $Q'_\star$ suggest the signal may instead arise from a late travel-time effect due to an unresolved companion at ~ 50 au. Together, these two systems illustrate the potential and limitations of transit timing in diagnosing tidal decay.



REFERENCES



J. N. Winn and G. Stef´ansson, (2025), arXiv:2510.05229.

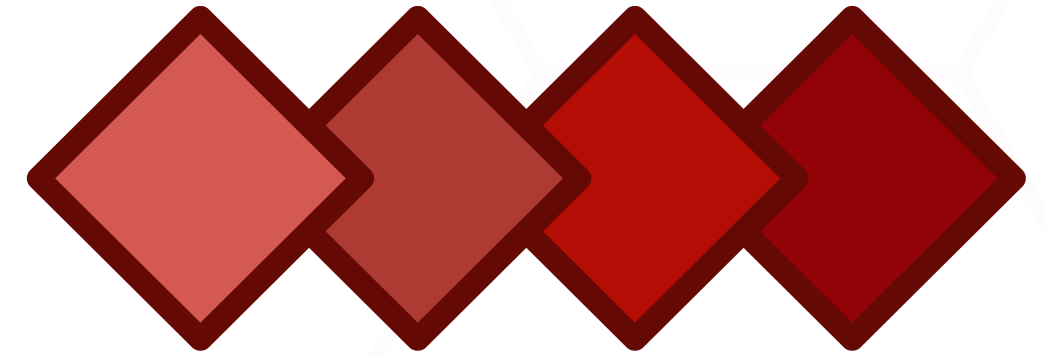
S. W. Yee et al., The Astrophysical Journal Letters 888, L5 (2019).

G. I. Ogilvie, Annual Review of Astronomy and Astrophysics 52, 171–210 (2014).

A. Chontos et al., The Astronomical Journal 157, 192 (2019).

S. Vissapragada et al., The Astrophysical Journal Letters 941, L31 (2022).

I. Wong et al., The Astronomical Journal 163, 175 (2022)



THANK YOU

